



Temperature shocks and energy poverty: Findings from Vietnam

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ABSTRACT

Globally, billions of people live in energy poverty without the means to access affordable and clean energy which are fundamental to sustainable development. Climate change is aggravating increasing demand for energy through a general rise in temperatures as well as through short periods of unusually high heat. This paper examines the impact of temperature shocks on energy poverty in Vietnam. Using data from the Vietnam Household Living Standards Survey (2010 to 2016) and data from the European Centre for Medium-Range Weather Forecasts it finds temperature shocks lead to increases in a measure of multidimensional energy poverty. This finding is robust to different measures of temperature shocks. Households in the North and Central coast regions of Vietnam are most prone to the impacts of temperature shocks as well as households located in rural areas. Further findings suggest that temperature shocks impact on energy poverty through reducing agricultural output.

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1. Introduction

Higher temperatures have been shown to have severe negative impacts on the well-being of households. For example, the prevalence of more frequent and intense heat waves has been linked to increased rates of mortality and morbidity (Tan et al., 2007; McMichael et al., 2008; Chung et al., 2009; Deschênes and Greenstone, 2011). It has also been associated with increased poverty (Hertel et al., 2010; Skoufias et al., 2011), food insecurity (Su et al., 2009; Rosegrant, 2011; Ziv et al., 2012) and other dimensions of poor health (Munslow and O'Dempsey, 2010).

The occurrence of higher temperatures is projected to increase in the coming decades (Christensen et al., 2007; Field et al., 2012). In response, most countries have accepted a legally binding international treaty to limit global warming to well below 2 and preferably 1.5 degrees Celsius (UNFCCC, 2015). Stemming the disruptions and displacements from climate change also requires more scientific evidence about its varied impacts, particularly on outcomes that have been overlooked in the past and on households whose livelihoods are sensitive to temperature changes.

In this paper, we assess the potential impact of higher temperatures on the outcome of energy poverty for individual households in Vietnam. Energy poverty refers to the concept of routinely having limited access

to and/or lack of ability to afford clean energy sources for conducting day-to-day activities, such as cooking or heating.¹ To our knowledge, there is little research examining the degree to which temperature changes may underlie the experience of energy poverty. A-priori, we expect temperature extremes to limit access to clean energy sources by reducing household income derived from agriculture-based production and by increasing total energy consumption, which under a constant budget constraint can lead to a substitution away from cleaner, more expensive energy sources to cheaper yet relatively unclean energy sources. Given these likely channels of effect and as temperature extremes are projected to become more frequent, having a greater understanding of their potential impact on energy poverty, is urgently needed.

A key motivation for focusing on energy poverty is that this form of deprivation, which Srivastava et al. (2012) described as one of the biggest social problems of the twenty-first century, continues to affect the lives of many people in developing countries and across some parts of Europe (Karpinska and Śmiech, 2020). For example, van der Kroon et al. (2013) estimate that two billion people continue to rely on traditional (biomass) fuels such as wood, agricultural residues and dung for their daily energy needs and about 1.4 billion people still suffer from a complete lack of access to electricity (Bridge et al., 2016). Apart

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¹ Note that the definition of energy poverty in developed countries refers to an inability to adequately heat their homes or to spend more than 10% of their income on energy expenditures. In developing countries, the concept of energy poverty revolves heavily around access and the ability to afford clean energy sources.

from its prevalence, this form of disadvantage begets further disadvantage for individuals, society and the environment. Traditional forms of energy, such as biomass or solid fuels, are often harvested unsustainably, often burned in inefficient cookstoves, and they emit harmful greenhouse gases, including black carbon and carbon dioxide (Sagar and Kartha, 2007; Kandlikar et al., 2009; Grieshop et al., 2011). The inefficiencies and durable environmental harms associated with such traditional sources imply that they are strictly inferior substitutes to modern energy sources.

Consequently, energy poverty has been linked to poorer health (Wilkinson et al., 2009; Recalde et al., 2019; Baudu et al., 2020), lower subjective well-being (Awaworyi Churchill et al., 2020), lower economic growth and greater environmental degradation (Kaygusuz, 2012; Nussbaumer et al., 2012; Araújo, 2014; Mainali et al., 2014; Halkos et al., 2015; Halkos, 2017; Lacey et al., 2017; Managi and Kumar, 2018).

Policies that stem the impacts of temperature extremes on energy poverty can thus have benefits that reach beyond alleviating immediate consumption deprivation – they can also fulfil development and environmental-sustainability goals. Reflecting this, signatories of the Sustainable Development Goals (SDGs) have adopted universal access to affordable and clean energy as one of the specific goals to achieve by 2030 (United Nations, 2015). Thus, by focusing on energy poverty as the outcome of interest, this paper makes a notable contribution to the climate-change literature, as well as to furthering the environmental cause.

For our analysis, we use an incredibly rich and nationally representative household survey dataset – the Vietnam Household Living Standards Survey (VHLSS). These data are uniquely suited to our aim as they allow us to discern detailed energy-use patterns for Vietnamese households. From this survey, we construct a multidimensional energy poverty index (MEPI), which accounts for deprivation in both the type and amount of energy used. We then link these survey data to high-frequency records of precise temperature data. Following the literature that documents non-linear impacts of temperature changes, and in particular, the dominant effects of heat stress (Park et al., 2010; Deschênes and Greenstone, 2011; Graff Zivin et al., 2018; Graff Zivin et al., 2020), we develop a variable for temperature shocks that is measured as a two (or more) standard deviation from a long-run mean temperature within a district in a specific year.

Our main approach is to estimate a linear probability model with a variety of fixed effects. For identification, our analysis uses the random variation of temperature shocks occurring across geography and time (households experiencing temperature shocks at different points in time). The attraction of exploiting both these sources of variation is that factors which may simultaneously influence the choice of residency and the likelihood of being energy poor are taken into account. Specifically, potential residential selectivity is likely to be common across households from the same province who differ in their exposure to a contemporaneous temperature shock simply because of the timing (or year) in which the shock takes place. Furthermore, as we compare the timing of the shocks over a short time frame (2010 to 2016), potential bias from differential macroeconomic trends that vary across provinces, is minimised. Our central identifying assumption is that temperature realisations are as good as random after accounting for spatial and temporal fixed effects, as well as province-linear time interactions. The benefit of our approach is that the impact of the temperature shock on energy poverty is better isolated.

Our main analysis uncovers a substantial effect of short-term deviations in temperature on energy poverty. Our results suggest that a positive-signed temperature shock (of two or more standard deviations from a district-specific long-run mean) will increase energy poverty by 2.5 percentage points based on the MEPI status and by 0.012 points based on the MEPI score. To put this into perspective, in our sample, the overall energy-poverty rate in Vietnam is 19% and 0.188 for MEPI status and MEPI index, respectively.

Consistent with existing studies on temperature susceptibility, we find the effects vary by region (Nguyen et al., 2007; World Bank, 2010). In particular, agricultural production in the North of Vietnam is particularly susceptible to the temperature shock impacts. The climate and topography that characterises the North is one of high elevation and relatively severe winters, which means most agricultural activity is concentrated in the lowland areas of the Red River Delta. For this reason, the rice yields in this region are highly sensitive to temperature changes (Trinh et al., 2020). The high level of irrigation applied to farms in this region is also a contributing factor (Trinh, 2018).

The remainder of this paper is structured as follows: Section 2 provides a brief review of the related literature and outlines our unique contributions to these bodies of work. Section 3 provides some background to the Vietnamese context; Section 4 outlines the data; Section 5 describes the methodology; Section 6 presents and discusses the results; and Section 7 concludes.

2. Review of the literature

Our research question relates to two key bodies of work. The first is the one that assesses the impact of temperature shocks on the broader experience of income poverty. A key similarity in studying the impact of temperature extremes on various facets of poverty, such as deprivation in energy, food, health and sanitation, is that a dominant mechanism through which the temperature effects are likely to be transmitted is that of reduced household income. Lower income, in turn, limits a household's ability to afford and access goods and services, including modern energy sources.

More specifically, temperature extremes constrain household income by reducing crop yields (Schlenker and Roberts, 2009; Feng et al., 2010; Welch et al., 2010; Fisher et al., 2012) and severely impacting on fish productivity and biodiversity (through higher sea levels and salinity intrusion) (Allison et al., 2009; Halls, 2009). Another mechanism through which temperature extremes can affect agriculture output is by triggering other climate events, such as floods and droughts, which can in turn deplete agricultural output (Trinh, 2018). Thus, the impact of temperature extremes is likely to be particularly strong for households whose livelihoods heavily depend on agricultural production.

All households, however, even those that are not agriculture-dependent, can experience reduced income from extreme temperatures. This is because temperature extremes dampen labour productivity and reduce labour supply (Huntington, 1924; Seppanen et al., 2003; Lee et al., 2014; Heyes and Saberian, 2019). This relationship has been documented for individuals from both developed and developing countries. Temperature extremes are likely to dampen labour productivity through its influence on mental or cognitive acuity (Hancock and Vasmatazidis, 2003; Fang et al., 2004; Hedge, 2004; Seppanen et al., 2006; Park et al., 2010; Graff Zivin et al., 2018). In general, the literature finds that it is heat stress, as opposed to cold stress, that has the greatest negative impact on the performance of cognitive tasks. The finding of the detrimental influence of high temperatures is a key motivation for our focus on higher temperatures.

Higher temperatures can also link to energy poverty in ways that are not directly related to income or labour productivity. Namely, more frequent temperature extremes can exert a direct increase in the demand for modern energy appliances, particularly, air-conditioners (Barreca et al., 2016). For households who have air conditioners and who are able to meet their higher energy needs, expenditures on energy as a proportion of total household income increases. This works to heighten the likelihood of experiencing energy poverty. In cases where households are unable to meet their cooling needs (because they lack access to or are unable to afford modern energy services) they may be more likely to suffer the consequences of reduced labour productivity. This is because workers, through the assistance of air conditioners, have been shown to successfully attenuate the dampening effects of extreme

temperatures on labour productivity (Dell et al., 2014). For energy-poor households, temperature extremes then may serve to reinforce their pre-existing disadvantage, compared to households with access to modern energy services.²

The second literature that we speak to is the one that looks specifically at the impacts of temperature extremes on energy consumption (Auffhammer and Aroonruengsawat, 2011; Deschênes and Greenstone, 2011; Loayza et al., 2012). This literature finds that stronger energy demand and consumption is provoked predominantly by extremely warm temperatures (Auffhammer and Aroonruengsawat, 2011; Deschênes and Greenstone, 2011).

We differentiate our paper from these bodies of work in two main ways. First, most of the literature focuses on developed countries. Much less is known about the temperature effects on energy demand or consumption outside of rich countries. We explore the energy poverty-higher temperature link for a country that is particularly vulnerable to both phenomena: Vietnam. For example, it is among the top five countries predicted to be adversely affected by natural disasters and climate change, owing to its topography, population size, and the concentration of economic activity in coastal areas (World Bank, 2010). Furthermore, in certain parts of the country, such as in the Mekong Delta region, current temperatures are already at critical levels such that increasing heat stress further is likely to significantly damage the process of rice development and impact fisheries (Halls, 2009; Wassmann et al., 2009). Vietnam's pronounced sensitivity to climate impacts is further compounded by the fact that many of its households heavily rely on agricultural production as their main source of income. This is estimated to be as high as approximately 40% of all Vietnamese households (World Bank, 2016; Bangalore et al., 2018). Also, despite Vietnam's rapid economic transformation in previous decades, energy poverty has been highlighted as a major and persistent barrier to future economic growth for the country (Nguyen et al., 2019).

Second, we specifically focus on the concept of energy poverty, which is different to that of energy consumption. For example, energy poverty focuses on the type of energy use (traditional biomass and solid fuels versus modern energy sources) and values the substitution between these energy sources. While increased consumption of biomass or fossil fuel-based energy mix would be considered an 'improvement' according to the measure of energy consumption, it would be considered a 'worsening' of the experience of energy poverty. This distinction is important because policies that shape the type of energy use, as opposed to a focus on the intensity or affordability of energy, have significantly different implications for environmental and climate policies. Namely, policies that promote higher demand for energy, regardless of the type of energy use, potentially increase greenhouse gas emissions, which subsequently increase future energy demand. By contrast, policies that specifically target the substitution of traditional energy sources for modern energy sources can avoid this feedback loop dilemma. Thus, they can serve to fulfil environmental sustainability goals (Kaygusuz, 2012; Halkos et al., 2015; Trotta, 2020). For example, it is estimated that policies specifically aimed at eradicating energy poverty, such as erecting additional energy infrastructure, would produce minimal additional warming to the climate – an impact that is manageable considering the other benefits that a successful energy poverty reduction policy would bring (Allen and Barnes, 1985; Geist and Lambin, 2002; Hofstad et al., 2009; Köhlin et al., 2011).

A distinct benefit of the survey data used is that we can observe both the 'amount' of energy-use, as well as the 'type' of energy use. Furthermore, we can identify the types of energy used to service distinct purposes, such as for cooking, lighting, services, entertainment/education and communication. This means that we can construct a multidimensional energy poverty index. Our MEPI construction borrows from the

innovations made in the existing literature on the measurement of energy poverty – one that stresses the need to account for three aspects of energy deprivation: (1) the type of energy use; (2) the expenditure on and consumption of energy; and (3) the durable assets required to service day-to-day energy-related purposes (Nussbaumer et al., 2012; Ogwumike and Ozughalu, 2016; Sokołowski et al., 2019; Awaworyi Churchill and Smyth, 2020). The adoption of the MEPI also allows us to examine the incidence of energy-related hardship which captures whether or not households are energy poor (a binary measure that equals to one if the household's MEPI exceeds 0.33); and its severity, based on the average number of energy related indicators a household is deprived. We compare our findings using this preferred measure with alternative measures of energy poverty, which are detailed in Section 4.

3. The Vietnamese context

This section provides a brief background on the two issues in Vietnam that are central to this paper: the country's vulnerability to temperature shocks; and the incidence and nature of energy poverty at the household level.

Vietnam is located in South East Asia, sharing a border with China to the North and with Laos and Cambodia to the West. The country has four distinct geographic regions: the North; the Central coast; the Central highlands; and the South, which differ considerably in terms of latitude and therefore in climate and weather patterns. The North has greater seasonal variations in temperature. In summer, the North experiences average temperatures of between 22.5 and 27.5 °C compared to 15 to 20 °C in winter. This compares to average temperatures of 28 to 29 °C dropping to 26 to 27 °C for these seasons in the South (McSweeney et al., 2010). The Central coastal provinces experience hot and dry summers as well as distinct rainy seasons while in the Central highlands the temperature tends to be much cooler. Fig. 1 provides mean temperatures by district over the period 1980–2009.

Vietnam is one of the most hazard-prone countries of Asia-Pacific Region and has been ranked among the five countries likely to be most affected by climate change (World Bank and Asian Development Bank, 2020). Mean annual temperature has increased by 0.4 °C since 1960 and this warming has been more rapid in the southern region of Vietnam. Importantly, the frequency of hot days and hot nights has increased significantly since 1960 in every season while the frequency of cold days and nights has fallen (McSweeney et al., 2010).³

According to General Circulation Models (GCM), the mean annual temperature in Vietnam is projected to increase by 0.8 to 2.7 °C by the 2060s, and 1.4 to 4.2 degrees by the 2090s (McSweeney et al., 2010). The greatest increases in temperature are projected to occur in the South of the country, followed by the Central regions. The northern region is expected to experience the smallest temperature increases. Variations in temperatures are also increasing. The rises in annual maximum and minimum temperatures are expected to be stronger than the rise in average temperature which will exacerbate the impacts of climate change (World Bank and Asian Development Bank, 2020). Fig. 2 plots temperature deviation (measured as the difference between actual temperature and its long-run mean, divided by the long-run standard deviation) over the period 1980 to 2020. The figure confirms large annual deviations in temperature for Vietnam. Further, the upward sloping trend line shows that temperature deviations in Vietnam have been increasing during the past four decades.

As discussed in Section 2, temperatures drive household demand for energy through the need for cooling and heating. Energy is also required for cooking, lighting and the use of modern appliances such as fridges and TVs. The demand for energy in Vietnam has increased rapidly in

² Recent evidence shows that extreme outdoor temperature can lead to poorer decision-making abilities despite having access to good-quality, climate-controlled indoor spaces (Heyes and Saberian, 2019).

³ Hot [cold] days or hot [cold] nights are defined as the temperature exceeded [falling below] on 10% of days or nights in the current climate of that region and season (McSweeney et al., 2010).

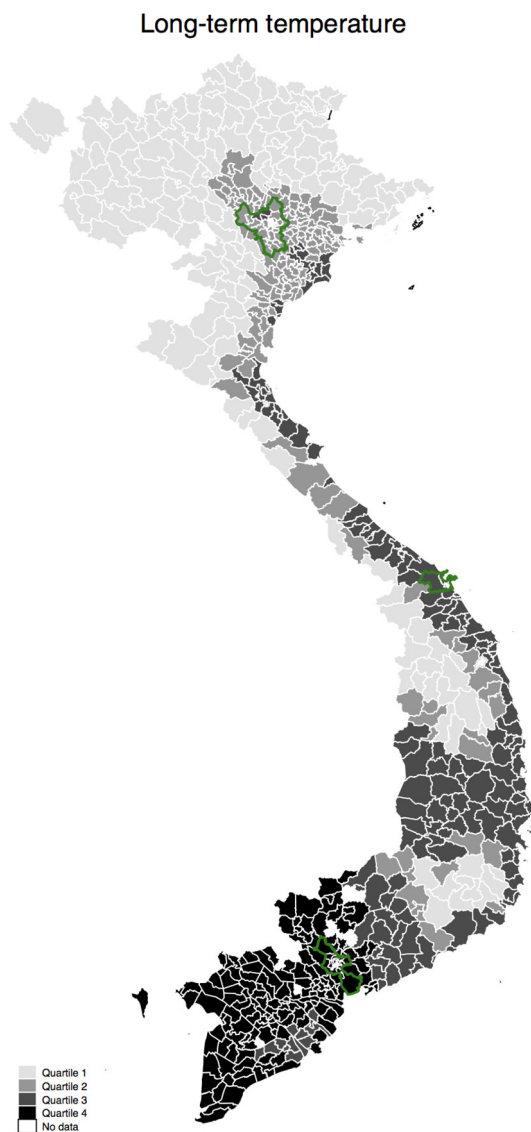


Fig. 1. Temperature across regions. *Notes:* Data from ERA5 database. Long-term temperature is averaged over the period 1980–2009. Darker values represent higher long-term temperature. Areas with black border represent three big cities in Vietnam (Hanoi, Da Nang, and Ho Chi Minh City).

recent decades due to the country's strong macroeconomic performance and sustained growth in household incomes. Since the *Doi Moi* economic reforms initiated in 1986, Vietnam has made considerable progress towards meeting the population's energy needs. Electricity generation has increased substantially in recent decades and household access to the national electricity grid in Vietnam is now almost universal (Ha-Duong and Nguyen, 2017; EVN, 2018). As a result of massive investments in rural electrification by the Vietnamese government and international aid donors, the proportion of the population with access to electricity increased from just 14% in 1990 to 97% by 2010 (World Bank, 2011). Electricity production increased ten-fold over this same period (Scott and Greenhill, 2014). Tang et al. (2016) show that energy consumption is an important determinant of Vietnam's economic growth.

The government of Vietnam has also adopted policies to ensure electricity is affordable. Between 1998 and 2009 it imposed a price ceiling on electricity for rural households. Since 2009, an electricity tariff has been applied which ensures that higher-income households, using

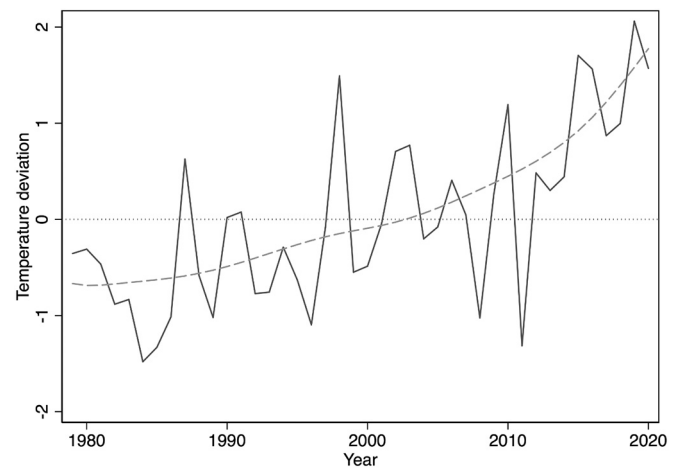


Fig. 2. Temperature deviation over time. *Notes:* Temperature deviation is measured as the difference between actual temperature and the long-run means, divided by the long-run standard deviation. The dashed line represents the Lowess curve.

more electricity, subsidise the cost of electricity for lower-income households (Scott and Greenhill, 2014).

However, there are still an estimated one million people without access to electricity, predominantly in the mountainous northern region of the country (Asian Development Bank, 2015). This is reflected by the graphs in Fig. 3 presenting the incidence of energy poverty according to our four different measures (defined and discussed in Section 4 below). They demonstrate that while there are pockets of energy poverty in the South and, to a greater extent, the Central region of Vietnam, the incidence is heavily concentrated in the North.

Having access to electricity does not imply that households can afford its consumption. For example, in 2010, one quarter of households in Vietnam reported that their electricity use was insufficient to meet their needs (Ha-Duong and Nguyen, 2017). Scott and Greenhill (2014) argue that rural households need to consume more than the subsidised amount of electricity (50 kWh a month) in order for electricity to significantly contribute to poverty reduction.

Further, as discussed energy poverty is not just concerned with access to electricity but also with the type of energy households consume. Less than two-thirds of the population in Vietnam rely primarily on clean fuels and technologies (WHO, 2020). Using data for the period 2004 to 2016, Nguyen et al. (2019) find that while households in Vietnam made a transition to modern energy, the poor and ethnic minorities continue to rely heavily on traditional forms of energy such as coal and biomass. They also find that despite electricity poverty decreasing during the period, the incidence of energy-cost poverty increased. Our data reflects a similar pattern. Fig. 4 plots our four different measures of energy poverty for our sample period from 2010 to 2016. The incidence of energy poverty according to all measures has fallen through time, most markedly for our preferred measure: the MEPI. The incidence of MEPI fell from 27% in 2010 to 15% in 2016. However, the incidence of energy poverty using the 10% measure barely changed from 2010 to 2014 before falling slightly in 2016.

4. Data

This study employs data from two sources. The first is the Vietnam Household Living Standard Survey (VHLSS). The VHLSS has been carried out biannually since 2000 by the General Statistics Office of Vietnam with technical support from the World Bank. It is a national representative sample for all provinces in Vietnam and each wave sample comprises approximately 9000 households. With the objective of measuring living standards across Vietnam, the data provide an enormous amount of household information including demographic

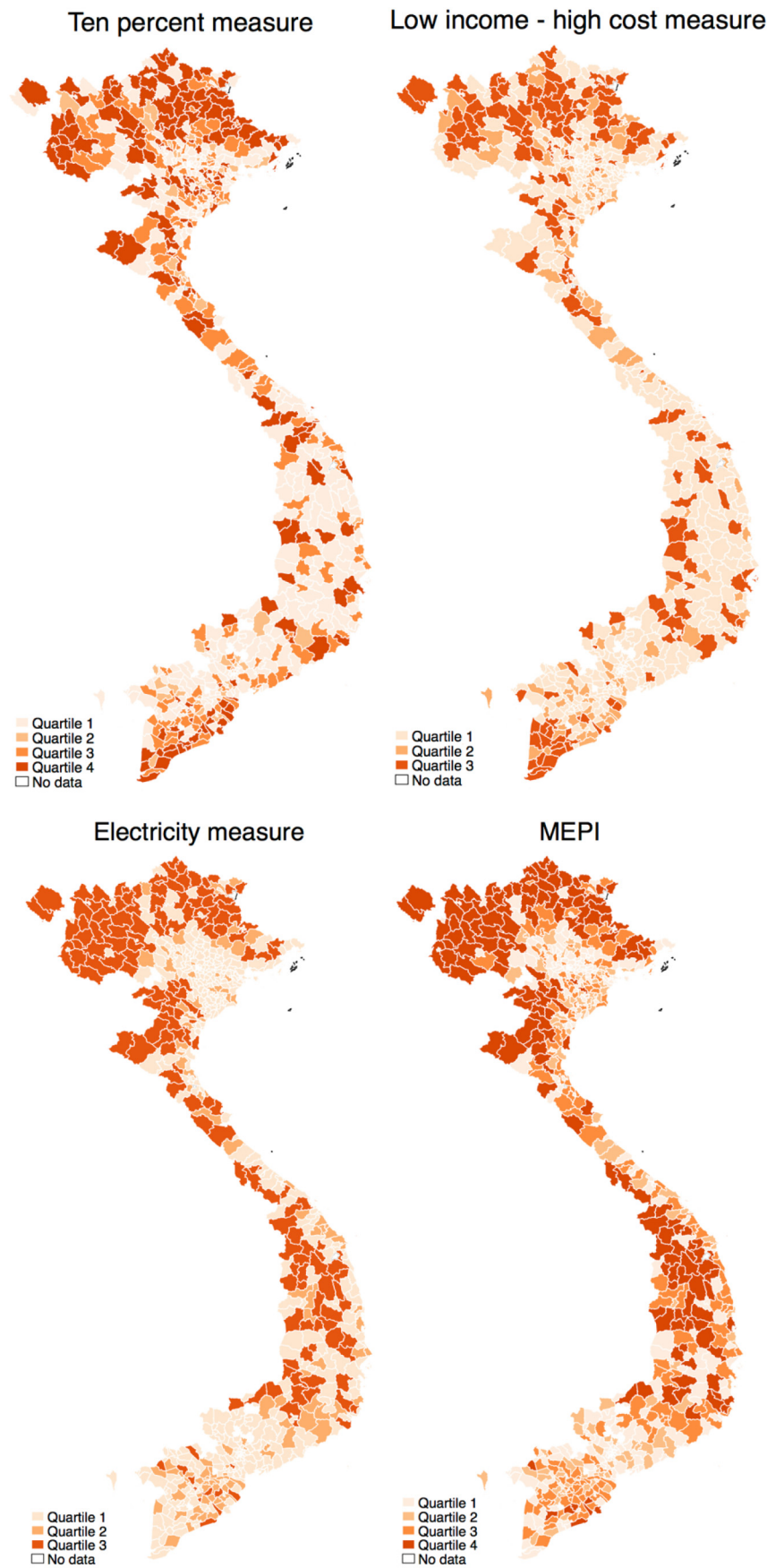


Fig. 3. Energy poverty across districts. Notes: Data from VHLSS 2016 at district level. Darker values represent higher rates of energy poverty.

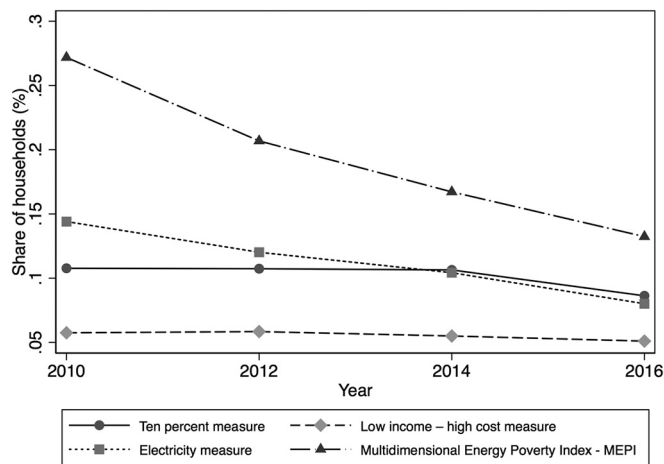


Fig. 4. Energy poverty over time. (Source: VHLSS 2010–2016.)

characteristics of household members, household income, expenditure, assets, housing and access to services. The survey relies on a rotating panel of households and is not designed for long panel data analysis (half of the households are randomly renewed from one round to the other). In our analysis, we focus on the VHLSS from 2010 to 2016. We do not employ the surveys before 2010 for two reasons. First, there was a significant change in data coding from 2010. Second, information on our main component of MEPI – electricity usage – is not available prior to 2010.

The household data are matched geographically with the global weather dataset taken from ERA5 reanalysis data, produced by the European Centre for Medium-Range Weather Forecasts (ECMWF). ERA5 is the fifth generation of ECMWF atmospheric reanalyses of the global climate. It provides high-quality reanalysis data using information from ground stations, satellites, weather balloons, and other inputs with a climate model to estimate weather variables across a grid (Dell et al., 2014). ERA5 provides data at a geographical resolution of 31 km and has been reformatted to a 0.25×0.25 degrees latitude-longitude grid. Currently, the weather data are available from January 1979 onward, with a temporal resolution of up to one hour. We draw on the daily mean temperature as well as total monthly precipitation and aggregate them at the district level, using the Google Earth Engine. Districts are the finest geographical level that we are able to identify households in Vietnam.⁴

4.1. Energy poverty

Although there is a substantial literature on the causes and effects of energy poverty, a consensus on the most appropriate way to measure energy poverty has yet to emerge. This is particularly true for developing countries where information on energy consumption is often limited. Among the most common approaches to measuring energy poverty are (i) economic based approaches (examining what proportion of income households need to spend on energy and how fuel costs compare to income), (ii) quantity based (such as whether a household's consumption of energy is below a specific threshold), (iii) subjective measures (where households are asked directly whether

they are able to meet their energy needs), and (iv) a multidimensional energy poverty (where several indicators of energy use, spending and assets are combined into a composite index). We provide a brief overview of each approach and discuss its advantages/disadvantages in Table A1 (Appendix).

Our preferred measure in this study is the Multidimensional Energy Poverty Index (MEPI) (Nussbaumer et al., 2012; Sadath and Acharya, 2017; Zhang et al., 2019; Awaworyi Churchill and Smyth, 2020). As discussed in Section 2, energy poverty is best measured using information relating to the access to affordable and clean energy. The MEPI captures a set of energy deprivations at the household level. Based on previous studies and data availability for Vietnam, we identify seven indicators of energy poverty, which are grouped into five broad dimensions: (i) cooking, (ii) lighting, (iii) services, (iv) entertainment/education, and (v) communication. All five dimensions are assigned an equal weight of 0.2. Table A2 (Appendix) shows the details of the indicators, weight, and its construction. Unfortunately, almost half of the households in the VHLSS were unable to provide their consumption of electricity (in kW). We view energy consumption as a vital component of energy poverty which must be included in our models. However, in Section 6 below, we examine whether this introduces sample selection bias to our estimations and whether our findings are sensitive to the exclusion of electricity consumption.

Each indicator is binary, taking the value of 0 or 1 depending on whether or not a household is deprived with respect to the indicator. For example, for the 'cooking' indicator, a household is deemed deprived if it does not have access to clean sources of energy (i.e., using coal/coal briquette/firewood/farm products). While this information was not sought directly from the interviewers for the VHLSS, we use information on household spending on cooking fuel as a proxy. Similarly, for the 'lighting' indicator, a household with per capita consumption of electricity of less than 100 kW per year is identified as deprived. We also employ information on durable assets to identify energy poverty including the ownership of a fridge, water heater, air conditioner, TV, and phone. The MEPI is calculated by summing the weighted number of deprivations. A household is then classified as multidimensionally energy poor if the sum of the weighted deprivations exceeds 0.33. This provides a household's MEPI status taking the value of one if poor and zero otherwise. We also calculate an MEPI score by multiplying MEPI status with the number of weighted indicators in which a household is deprived. The MEPI score therefore provides a measure of the depth of energy poverty for those households that are deemed multidimensionally energy poor.⁵ This is the approach undertaken by Alkire and Foster (2011) in calculating the depth of poverty in their Multidimensional Poverty Index on which subsequent MEPIs are based.

4.2. Temperature shocks

Previous studies have proposed several measures of temperature shocks. In this analysis, we identify the shock using temperature deviations. The deviations are calculated as the difference between actual temperature and the long-run average within a district in a specific year, divided by the long-run standard deviation for the same district. This approach has been widely used in the literature (e.g., Hirvonen, 2016; Letta et al., 2018; Graff Zivin et al., 2020). Historical temperature is measured from 1980 to 2009 which is prior to the time we measure energy poverty. Our main measure of a temperature shock is therefore a dummy variable equal to one if the temperature deviates by more than two standard deviations from its long run average.

We have defined our temperature shock variable in this manner for a number of reasons. First, we argue that level changes matter not in an absolute sense but rather in proportion to the district's usual variation.

⁴ Previous studies in Vietnam employ weather data from the gridded dataset provided by the Centre for Climatic Research, University of Delaware (e.g., Thai and Falaris, 2014; Trinh, 2020). Gridded datasets apply another approach to deal with missing weather observations for observationally sparse regions, namely interpolation of the existing weather information across space and time over a grid. For areas with sparse station coverage, weather data is interpolated from stations further away. This is problematic as the weather events that identify the response coefficient might vary according to the interpolation scheme and hence across datasets. The reanalysis data is therefore the preferred choice in this analysis.

⁵ Households deemed multidimensionally energy poor have a status value of one. This is then multiplied by the number (count) of weighted indicators in which the household is deprived. Thus the higher the number of deprivations, the higher the MEPI score.

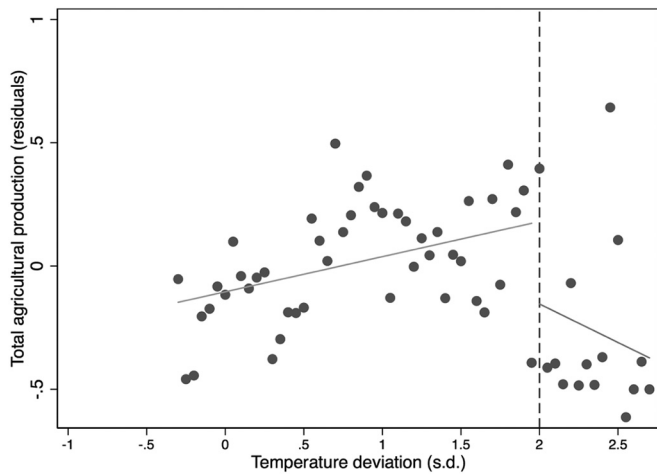


Fig. 5. Agricultural production and temperature shocks. *Notes:* The figure plots the results of regression with natural log of total agricultural as the dependent variable. The regression specification includes province and time fixed effects, interaction between province and a linear time trend, and survey month fixed effects.

Second, following the findings of previous studies, it is the occurrence of extreme temperatures that provokes much stronger responses to, for example, energy demand, labour productivity and reductions in crop yields (Auffhammer and Aroonruengsawat, 2011; Deschênes and Greenstone, 2011; Auffhammer and Mansur, 2014; Burke et al., 2015).

We show that temperature deviations above the two standard deviations threshold are likely to have an important impact on agricultural output. Fig. 5 plots the residuals from a regression of the log of agricultural production on province, time and survey month fixed effects against temperature deviations. It demonstrates that production tends to fall markedly when temperature deviations exceed two standard deviations. However, we also employ alternative measures of shocks using actual deviations (in levels) and different deviation thresholds as robustness tests.

By benchmarking the temperature change against a long-run average, we note that we implicitly account for adaptive responses. For example, extreme temperatures deviations (from a long-run average) that occur in South of Vietnam may have smaller effects because this part of the country routinely experiences hot climate. In response, residents may have already made investments in tolerating high temperatures e.g., buying air conditioners.⁶ This adaptation implies that our short-run estimate of a temperature shock may not be the same as the long-run impact of a permanent change in temperature of the same magnitude.

Furthermore, we calculate district-specific temperature shocks because different geographic areas within Vietnam have different baseline temperatures. Thus, an unusual weather shock in one district of the country may often be well within the normal experience in another district – indeed where adaptation has had the opportunity to occur. Therefore, the magnitude of the temperature change should be quantified relative to the district specific long-run temperature average.

5. Model specification

We are interested in estimating the causal effect of temperature shocks on energy poverty. We exploit quasi-exogenous variation in which households are exposed to a temperature shock. Our strategy is to compare energy poverty outcomes for residents living in districts of Vietnam that are equally likely to experience a temperature shock.

⁶ For example, in our data, approximately 87% of households in the South have air conditioners.

We estimate the following model specification:

$$EP_{ict} = \beta_0 + \beta_1 TS_{ct-1} + \theta' C_{ict} + \mu_p + \delta_t + \mu_p \times \lambda_t + \sigma_i + \varepsilon_{ict} \quad (1)$$

where EP_{ict} is the energy poverty status of household i in district c in year t (2010–2016). In the main specification, we employ all measures of energy poverty discussed above, and use the MEPI measure as our preferred proxy. The key explanatory variable is a temperature shock (TS_{ct-1}). We first calculate the difference between their average values in district c between year $t-1$ (the year preceding the measure of energy poverty) and the long-run district mean (1980–2009), divided by the long-run standard deviation. We therefore assume that level changes matter not only in an absolute sense but also in terms of their deviation from long-run averages. A temperature shock arises if the deviation is greater than two standard deviations from the long-term mean.

We control for a range of household and district characteristics, denoted in vector C_{ict} . Province fixed effects (μ_p) control for time invariant province level factors that influence energy poverty and time fixed effects (δ_t) account for any time trends in energy poverty that are common across provinces. We also include the interaction between province and a linear time trend ($\mu_p \times \lambda_t$) to control for province-specific macroeconomic trends. Survey month fixed effects (σ_i) are also included to control for the time of year the household was interviewed. The error term is denoted by ε_{ict} . Standard errors are robust to heterogeneity and are clustered at the district level.

The key assumption of Eq. (1) is that temperature realisations are as good as random after accounting for spatial and temporal fixed effects, as well as province-linear time interactions. Of course, there is no guarantee that the trends across an affected province, had a temperature shock not occurred, would exhibit a similar time-series pattern to a non-affected province. For example, we know that Vietnam experienced rapid economic growth during the last few decades and that this growth pattern was uneven across provinces. If residents sort based on these economic growth characteristics such that more vulnerable residents are more likely to live in provinces that are more subject to temperature shocks, then this would lead to an overestimate of β_1 . We have tried to minimise this issue in two ways. First, we control for linear time-province fixed effects and second, our analysis focuses on a relatively short time frame: from 2010 to 2016.

Table A3 (Appendix) provides summary statistics for the households variables in the VHLSS. On average, the share of energy-poor households in our sample is 19.5% using the MEPI approach. Males account for 75% of household heads in the period 2010–2016, and the average age of the household head is 50. The average household size is almost four persons per household, and approximately 50% of the households have a reinforced concrete roof.

6. Results

Our results section is comprised of four parts. We start by presenting our main results before examining whether there is heterogeneity in the temperature shock-energy poverty relationship. We proceed by undertaking several robustness tests and finally examining the mechanisms through which temperature shocks might impact on energy poverty.

6.1. Main results

Results from the estimation of Eq. (1) are presented in Table 1 below. For ease of exposition, only the results on the temperature shock variable, our key variable of interest, are presented. Full results are provided in Table A4 of the Appendix. The results suggest that while there is no relationship between temperature shocks and three of our measures of energy poverty (Columns 1–3), there is a positive and statistically significant impact of temperature shocks on the incidence and severity of multidimensional energy poverty (Columns 4–5). These results do

Table 1
Impacts of temperature shock on energy poverty.

Dependent variable:	Energy poverty				
	Ten percent measure	Low income – high cost measure	Electricity measure	MEPI status	MEPI score
	(1)	(2)	(3)	(4)	(5)
Temperature shock	0.002 (0.009)	0.002 (0.007)	0.008 (0.008)	0.025** (0.010)	0.012** (0.005)
Observations	36,238	36,238	36,238	36,238	36,238
R-squared	0.078	0.080	0.209	0.231	0.342
Other controls	Yes	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Province*linear time trend	Yes	Yes	Yes	Yes	Yes
Survey time FE	Yes	Yes	Yes	Yes	Yes

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$; Robust standard errors in parentheses; Standard errors are clustered at district level; Other controls include household head's characteristics (age, gender, educational level, household size, and type of rooftop) and district characteristics (average precipitation and population density). Full results are presented in Table A4 (Appendix).

not surprise us and justify the use of a multidimensional energy index. Temperature shocks will not increase the incidence of the 10% measure of energy poverty (households spending more than 10% of their income on energy) if shocks reduce household income but households use cheaper forms of energy. This is because household energy expenditures may fall in line with (or even more than) household income. This also applies to the 'Low income – high cost' measure which includes information on energy expenditures. Similarly, shocks will not necessarily increase the incidence of electricity consumption if households continue consuming the same amount of electricity but reduce spending on other items.

A temperature shock is associated with a 2.5 percentage point increase in the probability of being multidimensionally energy poor. It is also associated with a 1.2% increase in the severity of energy poverty, using the continuous measure of MEPI alone. As highlighted in the introduction, a potential explanation for this finding is that higher than usual temperatures result in higher household expenditures, particularly for electricity through cooling the household with an electric fan or air conditioner. Importantly, households may also have substituted clean, modern energy sources with cheaper, traditional sources, which move them into multidimensional energy poverty. It could also be the case that household income falls due to impacts on agricultural output, reduced labour supply or lower labour productivity (discussed further below). MEPI status and score are our preferred measures of energy poverty and we therefore focus on the results pertaining to these measures in the remainder of the paper.⁷

As discussed in Section 4, electricity consumption is only available for a sub-sample of households. To examine whether this leads to a sample selection issue, we estimate a model using a binary dependent variable, taking the value of one if household information on electricity consumption is missing and zero otherwise. We regress this variable on our temperature shock variable as well as the control variables specified in Eq. (1). Findings are presented in Table A6 of the Appendix and reveal that households lacking information on electricity consumption are no

⁷ To gain further insights we estimate the model separately for the different dimensions of the MEPI. Results are presented in Table A5 of the Appendix. They indicate that temperature shocks increase the probability of multidimensional energy poverty through the 'communications' component with households going without assets such as a phone. Taken with the results from Table 1, another interpretation of these findings is that it is the combination of the components which is important for multidimensional energy poverty rather than any individual component. This reinforces the need to construct a multidimensional measures of energy deprivation.

more (or less) likely to experience a temperature shock. To examine the issue further we omit electricity consumption from our MEPI and estimate our model on the full sample of households. Results are presented in Table A7 (Appendix) and are consistent with those presented in Table 1.

6.2. Examining heterogenous impacts

Thus far, we assume that temperature shocks have the same impact on multidimensional energy poverty for all households in Vietnam. However, temperature shocks might have different impacts on households in different geographic regions, based on their different weather patterns and other geographic factors, discussed in Section 3. We therefore estimate the model separately for the four distinct regions of Vietnam using both MEPI status (Panel A) and MEPI score (Panel B) as dependent variables. Results are presented in Table 2. Findings suggest that temperature shocks in the North and Central coast regions increase both the incidence and severity of multidimensional energy poverty and the impact is large. Temperature shocks calculated using deviations that are two standard deviations from the mean are associated with an increase in energy poverty exceeding 12 percentage points for households located in the North and over 4 percentage points for households located in the Central coast. This indicates that households in these regions are vulnerable to temperature shocks with fewer forms of resilience to cope with these shocks when they arise. There is also some evidence that temperature shocks defined in this way are associated with higher MEPI scores for households in the South but not with MEPI status. In other words, temperature shocks are associated with increases in the depth of multidimensional energy poverty but not its incidence in the South of Vietnam.

6.3. Robustness exercises

We conduct further robustness tests with respect to alternative measures of temperature shocks. Specifically, we employ different deviation thresholds at which temperature shocks are defined (2.5, 1.5, and 1 standard deviation(s) from the long-term average). We also examine the number of 'hot' days in a year defined as the number of days the temperature exceeds two standard deviations above its long-term average. Finally, we examine temperature deviation in levels (without specifying a shock threshold). Substantively, the results, which are

Table 2
Impacts of temperature shock on MEPI across regions.

	(1)	(2)	(3)	(4)
Regions	North	Central coast	Central highlands	South
<i>Panel A: Dependent variable is MEPI status</i>				
Temperature shock	0.123** (0.049)	0.043** (0.020)	−0.012 (0.039)	0.021 (0.022)
Observations	13,782	8138	2548	11,770
R-squared	0.329	0.170	0.317	0.138
<i>Panel B: Dependent variable is MEPI score</i>				
Temperature shock	0.053* (0.031)	0.032*** (0.010)	−0.032 (0.022)	0.021* (0.011)
Observations	13,782	8138	2548	11,770
R-squared	0.444	0.271	0.408	0.253
Average temperature	23.386	25.264	24.663	28.004
Other controls	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Province*linear time trend	Yes	Yes	Yes	Yes
Survey time FE	Yes	Yes	Yes	Yes

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$; Robust standard errors in parentheses; Standard errors are clustered at district level; Other controls include household head's characteristics (age, gender, educational level, household size, and type of rooftop) and district characteristics (average precipitation and population density).

Table 3
Robustness tests – alternative measures of temperature shock.

Dependent variable:	(1) MEPI status	(2) MEPI score
<i>Panel A: Cut-off = 2.5 standard deviations</i>		
Temperature shock	0.074*** (0.018)	0.056*** (0.011)
Observations	36,238	36,238
R-squared	0.231	0.343
<i>Panel B: Cut-off = 1.5 standard deviations</i>		
Temperature shock	0.016** (0.008)	0.010*** (0.004)
Observations	36,238	36,238
R-squared	0.231	0.342
<i>Panel C: Cut-off = 1 standard deviation</i>		
Temperature shock	0.007 (0.008)	0.004 (0.004)
Observations	36,238	36,238
R-squared	0.231	0.342
<i>Panel D: Number of days that are 2sd above long-run mean</i>		
Number of hot days	0.001*** (0.000)	0.001*** (0.000)
Observations	36,238	36,238
R-squared	0.231	0.343
<i>Panel E: Temperature deviation</i>		
Temperature deviation	0.023*** (0.009)	0.011** (0.004)
Observations	36,238	36,238
R-squared	0.231	0.342
Other controls	Yes	Yes
Province FE	Yes	Yes
Year FE	Yes	Yes
Province*linear time trend	Yes	Yes
Survey time FE	Yes	Yes

Notes: *** p < 0.01, ** p < 0.05, * p < 0.1; Robust standard errors in parentheses; Standard errors are clustered at district level; Other controls include household head's characteristics (age, gender, educational level, household size, and type of rooftop) and district characteristics (average precipitation and population density).

presented in Table 3, are remarkably consistent across the different measures. Temperature shocks of varying severity and form all lead to a higher MEPI status and MEPI scores. Shocks defined as temperature deviations exceeding one standard deviation from the long-term average have no impact. A one-standard deviation increase in temperature is common in Vietnam (occurring in 58% of periods), and as illustrated in Fig. 5, has little detrimental impact on agricultural production. This indicates that temperature shocks need to be of a certain intensity to impact on multidimensional energy poverty. Our results suggest that low intensity temperature shocks do not have an impact.

We examine further the impact of the number of hot days on multidimensional energy poverty by estimating this model by region. Results are presented in Table 4. Interestingly there is a positive association between the number of hot days and the incidence of multidimensional energy poverty in all four regions. However, the impact is small. There is also some evidence that the number of hot days impacts on the severity of poverty (MEPI score) in the Central coast and Central highlands regions.⁸

Arguably, it might take some time for a temperature shock to impact on household energy poverty. For example, households might be resilient to impacts in the short term but unable to buffer a reduction in income arising from the shock over a longer period. To examine this issue, we lag our shock variable by two and three periods and present the results in Table 5. Finding suggests that the impact of temperature shocks

Table 4
Number of hot days by region.

Regions	(1) North	(2) Central coast	(3) Central highlands	(4) South
<i>Panel A: Dependent variable is MEPI status</i>				
Number of hot days	0.003* (0.002)	0.002*** (0.001)	0.004*** (0.001)	0.001* (0.001)
Observations	13,782	8138	2548	11,770
R-squared	0.328	0.171	0.319	0.139
<i>Panel B: Dependent variable is MEPI score</i>				
Number of hot days	0.001 (0.001)	0.002*** (0.000)	0.003*** (0.001)	0.000 (0.000)
Observations	13,782	8138	2548	11,770
R-squared	0.443	0.273	0.410	0.253
Average number of 'hot' days	11.994	15.952	17.652	22.632
Other controls	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Province*linear time trend	Yes	Yes	Yes	Yes
Survey time FE	Yes	Yes	Yes	Yes

Notes: *** p < 0.01, ** p < 0.05, * p < 0.1; Robust standard errors in parentheses; Standard errors are clustered at district level; Other controls include household head's characteristics (age, gender, educational level, household size, and type of rooftop) and district characteristics (average precipitation and population density).

Table 5
Lagged impact of temperature shock on MEPI.

Dependent variable:	(1) MEPI status	(2) MEPI score
<i>Panel A: Shock measured in (t-2)</i>		
Temperature shock	0.013* (0.008)	0.009** (0.004)
Observations	36,238	36,238
R-squared	0.231	0.342
<i>Panel B: Shock measured in (t-3)</i>		
Temperature shock	−0.002 (0.008)	0.002 (0.004)
Observations	36,238	36,238
R-squared	0.231	0.342
Other controls	Yes	Yes
Province FE	Yes	Yes
Year FE	Yes	Yes
Province*linear time trend	Yes	Yes
Survey time FE	Yes	Yes

Notes: *** p < 0.01, ** p < 0.05, * p < 0.1; Robust standard errors in parentheses; Standard errors are clustered at district level; Other controls include household head's characteristics (age, gender, educational level, household size, and type of rooftop) and district characteristics (average precipitation and population density).

occurring two-year prior increases both MEPI status and score (but by small amounts). Shocks occurring three years ago do not appear to have an impact.

Next, we use alternative construction of MEPI to test the robustness of our results. In the main analysis, we assign equal weights to the dimensions of MEPI following Awaworyi Churchill and Smyth (2020). As an alternative, we attach a weight of 0.4 to the cooking dimension, 0.3 to the lighting dimension, 0.1 for services, entertainment/education, and 0.1 to communications. This is justified by cooking being among the most basic of needs (Alem and Demeke, 2020; Zhang et al., 2021). Traditional cooking appliances and fuels impose significant cost on households, both in terms of time and the negative health effects from indoor air pollution. Our results remain robust as shown in Panel A of Table A8 (Appendix). We also use a cut-off of 0.5 (instead of 0.33) to identify if a household is energy poor. The results presented in Panel B of Table A8 (Appendix) show that our findings remain consistent.

⁸ We also estimate a model which includes a quadratic term of the number of hot days variables, to examine whether there is a non-linear relationship between hot days and multidimensional energy poverty. Results provided no evidence of such a relationship.

The heterogeneity of our results is further explored in Table 6. We examine whether the impact of temperature shocks varies for households in rural versus urban areas as well as for the majority Kinh ethnic group versus ethnic minorities in Vietnam. This is achieved through the introduction of interaction terms. The results suggest that temperature shocks increase multidimensional poverty for rural households (both MEPI status and score) relative to urban ones. This is likely explained by rural households being dependent on agriculture for a living and temperature shocks having a greater impact on their incomes due to the disruption these shocks have on agricultural production. Impacts of temperature shocks do not appear to vary greatly across Kinh versus ethnic minority households although there is some evidence that the impact of shocks is less for Kinh households with respect to their MEPI scores. Impacts might be greater for these minority groups as they tend to be more concentrated in upland and mountainous areas, often with poor access to public services and basic infrastructure and therefore a higher vulnerability to income shocks (Van de Walle and Gunewardena, 2001; Baulch et al., 2007; Aroui et al., 2015). Our finding is also in line with studies in the energy literature which shows that different ethnic groups behave differently towards energy conservation (e.g., Awaworyi Churchill and Smyth, 2020; Wang et al., 2021).

6.4. Examining mechanisms

We proceed by explicitly examining the mechanisms through which temperature shocks might impact on multidimensional energy poverty. We postulate that the impacts are likely to arise through a loss of household income (measured using the log of total household income and income per capita), either due to a loss of productivity or through a loss in agricultural output (measured using the log of crop yield). We add these potential mediators, one at a time to our baseline regression using both the MEPI status and score variables. If they are indeed mediators which link temperature shocks to increase multidimensional energy poverty, we would observe a statistically significant coefficient on these variables together with an improved *R*-squared. Additionally, the coefficient on the temperature shock variable should be smaller since the negative impact from shock experience will be partially explained by the fall in household income and crop yield.

Results presented in Table 7, suggest reduced crop yields may mediate the fall in multidimensional energy poverty as a result of

Table 6
Heterogeneity tests.

Dependent variable	(1) MEPI status	(2) MEPI score
<i>Panel A: Urban vs rural</i>		
Temperature shock	0.008 (0.011)	0.002 (0.006)
Temperature shock*Rural	0.042*** (0.010)	0.024*** (0.006)
Observations	36,238	36,238
R-squared	0.159	0.252
<i>Panel B: Kinh vs minorities</i>		
Temperature shock	0.012 (0.020)	0.021* (0.012)
Temperature shock*Kinh	−0.000 (0.020)	−0.021* (0.011)
Observations	36,238	36,238
R-squared	0.210	0.301
Other controls	Yes	Yes
Province FE	Yes	Yes
Year FE	Yes	Yes
Province*linear time trend	Yes	Yes
Survey time FE	Yes	Yes

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$; Robust standard errors in parentheses; Standard errors are clustered at district level; Other controls include household head's characteristics (age, gender, educational level, household size, and type of rooftop) and district characteristics (average precipitation and population density).

Table 7
Temperature shock and energy poverty – mechanism analysis.

	Dependent variable: MEPI status			
	(1)	(2)	(3)	(4)
<i>Panel A: MEPI status</i>				
Temperature shock	0.022** (0.011)	0.023** (0.010)	0.024** (0.010)	0.027* (0.016)
Total income (in log)		−0.204*** (0.003)		
Income per capita (log)			−0.226*** (0.003)	
Crop yield (log)				−0.011*** (0.001)
Observations	36,238	36,238	36,238	17,555
R-squared	0.126	0.285	0.277	0.212
Province FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Province*linear time trend	Yes	Yes	Yes	Yes
Survey time FE	Yes	Yes	Yes	Yes
<i>Panel B: MEPI score</i>				
Temperature shock	0.010 (0.006)	0.010** (0.005)	0.011** (0.005)	0.008 (0.009)
Total income (in log)		−0.138*** (0.002)		
Income per capita (log)			−0.154*** (0.002)	
Crop yield (log)				−0.005*** (0.001)
Observations	36,238	36,238	36,238	17,555
R-squared	0.189	0.423	0.414	0.301
Province FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Province*linear time trend	Yes	Yes	Yes	Yes
Survey time FE	Yes	Yes	Yes	Yes

Notes: Robust standard errors in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$; Standard errors are clustered at district level.

temperature shocks. The coefficient on the mediator variables is negative and statistically significant while the coefficient on the temperature shock variable reduces to nearly zero in the case of the MEPI score and/or reduces in statistical significance for both the MEPI status indicator and the MEPI score. The adverse impact of a temperature shock on agricultural production has been confirmed in the literature. For example, warmer temperatures can exacerbate droughts during rainfall deficiencies due to higher evaporation and increased water use uptake by plants, thereby increasing total irrigation costs (Trinh, 2018). Higher temperatures also lead to lower labour productivity, especially in climate-exposed sectors such as agriculture (Graff Zivin and Neidell, 2014). Temperature shocks, therefore, can result in a higher incidence of energy poverty through lowering agricultural income. Surprisingly, we see little evidence that total income operates as a channel of effect. This may arise because the reported income is subject to large measurement error, especially as agricultural-dependent households conduct much of their economic activity in non-formal sectors.

7. Conclusion

We provide, firstly, evidence for the effect of a temperature shock on levels of multidimensional energy poverty. Using rich panel-based survey data combined with precise temperature information, we find that a significant contributor to multidimensional energy poverty in Vietnam is the incidence of higher temperatures.

A key insight provided by our results is that temperature shocks lead households to forego cleaner sources of energy or modern energy devices. Instead, households rely on traditional forms of energy in order to keep overall energy expenditure ratios (as a proportion of household

income) constant. This result highlights the need to focus on both the type and amount of energy consumed (rather than the latter alone) in order to elucidate the multi-faceted nature of energy deprivation.

Our core finding is that a temperature shock (of two or more standard deviations from a long-run average) increases the incidence of energy poverty by 2.5 percentage points based on a multidimensional energy poverty indicator. It also increases the severity of energy poverty by 1.2%. This finding is robust to variations in the thresholds defining temperature extremes, the use of different temperature intensity variables (such as the number of hot days), sampling decisions, and other model specifications.

Interrogating these results by different geographic regions within Vietnam, we find that the impact of higher temperatures on multidimensional energy poverty incidence and severity is concentrated in the Northern and Central coast regions. This aligns with previous findings showing that these regions are more sensitive to climate change impacts. Furthermore, we find rural households, as opposed to urban households, experienced more pronounced impacts. This is unsurprising given that rural households depend on agriculture for a living and temperature shocks directly impact on agricultural production.

This finding is substantiated in our mechanism analysis which provides suggestive evidence that a reduction in crop yields is a potential pathway through which higher temperatures affect energy poverty. Potential reporting errors in household income prevent us from pinning down income as a mechanism from higher temperatures to energy poverty. Using broader measures of income and wealth would be a useful way to build on this mechanism analysis. Unfortunately, this was not available in our survey.

The findings from this paper are likely to apply to many other countries, particularly those located in South East Asia. They highlight a number of considerations for policymakers. Given the link between climate change and temperature shocks the international community should strengthen its commitment to reduce carbon dioxide emissions in order to curb future temperature rises.

At a national level, two broad policy recommendations apply. Firstly, targeted policy interventions are likely to be most efficient. The incidence and depth of multidimensional energy poverty is highest for

households in the North and Central coast regions of Vietnam. They are also most prone to experiencing the impacts of temperature shocks. It is households in these regions to which assistance should be targeted.

Secondly, as noted in Section 3, while Vietnam has made considerable progress at providing households access to electricity this does not imply that households can afford its consumption. The incidence of energy-cost poverty remains high in Vietnam and some households still rely heavily on traditional forms of energy such as coal and biomass. Moreover, our findings suggest that temperature shocks lead households to forego cleaner sources of energy or modern energy devices. To reduce multidimensional energy poverty, it is therefore necessary to reduce the cost of electricity for poor households. This could be achieved through the provision of subsidies or cash transfers to energy poor households, possibly supported by more progressive electricity tariffs (Ha-Duong and Nguyen, 2017). It could also be achieved by improving the efficiency of electricity production. Currently, virtually all electricity in Vietnam is produced by State Owned Enterprises. Allowing greater private investment and competition in this sector is likely to lead to the cheaper provision of electricity and therefore to reductions in energy poverty. Improving efficiency in production could complement low-cost interventions aimed at improving household energy efficiency (such as insulation, sealing doors and windows against drafts and the use of blinds and energy efficient appliances) which will increase the resilience of households to temperature shocks.

Finally, understanding the varied impacts of higher temperatures is crucial to forming a measured view of the implications of climate change. World average temperatures are rising as are the frequency of very hot days. That higher temperatures can make households more energy deprived in the short-term is only the first step to the climate-change impact chain. Lacking access to clean and affordable sources of energy can subsequently lead to worse-off health, lower incomes, and greater environmental degradation. The long-term and knock-on effects of climate change, and the mechanisms underlying them, are important avenues of ongoing research.

Appendix

Table A1
Common measures of energy poverty.

Type, name and source of measure	Definition	Advantages	Disadvantages
Income and expenditure based			
The 10% measure (Boardman, 1991)	Households needing to spend more than 10% of the income to pay the energy bill (including electricity and fuel for heating and cooking excluding transportation).	Data widely available. Easy to calculate and understand.	Errors in the reporting of income and energy expenditures. Rich households consuming a lot of energy could be deemed poor. Amount needed to spend on energy is driven by the energy efficiency of the dwelling.
The low income – high cost measure (Hills, 2012)	Poor if fuel costs are above the median level and if residual income falls below the official poverty line (or per capita household disposable income is smaller than 60% of the national median income).	Better captures low-income households which have high energy requirements.	Complex methodology to estimate energy costs, to which two arbitrary thresholds are applied (median costs and a subjective poverty line).
Quantity based			
Electricity poverty (Coelho and Goldemberg, 2013)	A household is considered to be electricity poor if its per capita consumption of electricity is less than 100 kWh per year.	It is ultimately the quantity of energy consumed by a household that is important rather than what they spend on energy costs.	A household might reach this threshold but only by spending a large proportion of its total income.
Subjective			
Subjective energy poverty measure (Waddams Price et al., 2012)	Households asked directly whether they feel they can afford their energy needs.	Households are best placed to ascertain whether their energy needs are being met.	Respondents can have very different expectations/standards and interpretations of energy needs.
Multidimensional			
Composite energy poverty indices (Nussbaumer et al., 2012)	Typically includes income, energy expenditure, types of fuel consumed, ownership of a fridge, TV, computer.	Recognises that there are multiple dimensions to energy poverty. Captures both the incidence and intensity of energy poverty.	Arbitrary selection of dimensions, indicators, deprivation thresholds, weights and method for aggregating the components. Dimensions will differ across countries according to climate and customs.

Table A2
Multidimensional Energy Poverty Index.

Dimension (weight)	Indicator (weight)	Deprivation (poor) if...
Cooking (0.2)	Modern cooking fuel (0.1)	Use coal/coal briquette/firewood/farm products (based on expenditure)
Lighting (0.2)	Indoor pollution (0.1)	Do not have gas/magnetic/electric cooker
Services (0.2)	Electricity usage (0.2)	Per capita consumption of electricity less than 100 kW per year
	Household appliance ownership (0.1)	Has no fridge
	Heating or cooling (0.1)	No water heater and no electric fan and no air conditioner
Entertainment/Education (0.2)	Household appliance ownership (0.2)	Has no TV?
Communication (0.2)	Telecommunication means (0.2)	Has no phone

Table A3
Summary statistics.

Variable	Description	Mean	St. Dev.
Energy poverty measures			
Income-based	Poor if households spend more than 10% of the income to pay the energy bill.	0.100	0.301
Poverty-based	Poor if fuel costs are above the median level and if residual income falls below the official poverty line.	0.055	0.228
Electricity-based	Poor if per capita consumption of electricity is less than 100 kWh per year.	0.111	0.315
MEPI dummy	Multidimensional energy poverty index (dummy)	0.195	0.396
MEPI	Multidimensional energy poverty index	0.189	0.220
Household and district characteristics			
Age	Age in years	50.305	14.120
Gender	Male = 1	0.748	0.434
Education	Years of education	7.303	3.689
Household size	Number of household members	3.855	1.564
Roof material	Reinforcement concrete = 1	0.497	0.500
Population density	Population density at district level	0.193	0.580
Rainfall	Average rainfall at district level (metre)	0.006	0.004
Temperature shock			
Temperature shock (2sd)	= 1 if temperature deviation >2 standard deviations	0.133	0.339
Temperature shock (1sd)	= 1 if temperature deviation >1 standard deviation	0.585	0.493
Temperature shock (1.5sd)	= 1 if temperature deviation >1.5 standard deviations	0.299	0.458
Temperature shock (2.5sd)	= 1 if temperature deviation >2.5 standard deviations	0.015	0.123
Number of hot days	Number of days that are 2SD above long-run mean	17.745	10.652

Table A4
Impacts of temperature shock on energy poverty – Full results.

	Dependent variable: Energy poverty				
	Ten percent measure	Low income – high cost measure	Electricity measure	MEPI status	MEPI score
	(1)	(2)	(3)	(4)	(5)
Temperature shock	0.002 (0.009)	0.002 (0.007)	0.008 (0.008)	0.025** (0.010)	0.012** (0.005)
Household size	−0.023*** (0.001)	0.021*** (0.001)	0.026*** (0.001)	−0.012*** (0.002)	−0.009*** (0.001)
Gender (Male = 1)	0.005 (0.004)	0.012*** (0.002)	0.011*** (0.004)	0.000 (0.005)	0.005* (0.003)
Age	0.001*** (0.000)	−0.000*** (0.000)	−0.002*** (0.000)	−0.001*** (0.000)	−0.001*** (0.000)
Roof material (concrete = 1)	−0.017*** (0.005)	−0.003 (0.004)	−0.060*** (0.005)	−0.075*** (0.006)	−0.046*** (0.003)
Education	−0.010*** (0.001)	−0.006*** (0.000)	−0.021*** (0.001)	−0.036*** (0.001)	−0.024*** (0.000)
Population density	0.004 (0.003)	−0.003*** (0.001)	0.002 (0.002)	−0.007** (0.003)	−0.013*** (0.002)
Rainfall	0.178 (0.540)	0.723* (0.403)	−0.537 (0.495)	−0.079 (0.642)	−0.129 (0.344)
Observations	36,238	36,238	36,238	36,238	36,238
R-squared	0.078	0.080	0.209	0.231	0.342
Province FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Province*linear time trend	Yes	Yes	Yes	Yes	Yes
Survey time FE	Yes	Yes	Yes	Yes	Yes

Notes: *** p < 0.01, ** p < 0.05, * p < 0.1; Robust standard errors in parentheses; Standard errors are clustered at district level.

Table A5

Impacts of temperature shock on components of MEPI.

	Dependent variable: Energy poverty				
	Cooking	Electricity usage	Services	Entertainment/ Education	Communication
	(1)	(2)	(3)	(4)	(5)
Temperature shock	0.010 (0.009)	0.008 (0.008)	0.011 (0.009)	0.002 (0.008)	0.033*** (0.009)
Observations	36,238	36,238	36,238	36,238	36,238
R-squared	0.231	0.209	0.178	0.082	0.170
Other controls	Yes	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Province*linear time trend	Yes	Yes	Yes	Yes	Yes
Survey time FE	Yes	Yes	Yes	Yes	Yes

Notes: *** p < 0.01, ** p < 0.05, * p < 0.1; Robust standard errors in parentheses; Standard errors are clustered at district level; Other controls include household head's characteristics (age, gender, educational level, household size, and type of rooftop) and district characteristics (average precipitation and population density).

Table A6

Sample bias test.

Dependent variable	(1)
	Missing electricity consumption = 1
Temperature shock	−0.004 (0.007)
Observations	73,330
R-squared	0.572
Other controls	Yes
Province FE	Yes
Year FE	Yes
Province*linear time trend	Yes
Survey time FE	Yes

Notes: *** p < 0.01, ** p < 0.05, * p < 0.1; Robust standard errors in parentheses; Standard errors are clustered at district level; Other controls include household head's characteristics (age, gender, educational level, household size, and type of rooftop) and district characteristics (average precipitation and population density).

Table A7

Impacts of temperature shock on energy poverty – Full sample.

Dependent variable	(1)	(2)
	MEPI status	MEPI score
Temperature shock	0.031*** (0.007)	0.020*** (0.004)
Observations	73,330	73,330
R-squared	0.240	0.358
Other controls	Yes	Yes
Province FE	Yes	Yes
Year FE	Yes	Yes
Province*linear time trend	Yes	Yes
Survey time FE	Yes	Yes

Notes: *** p < 0.01, ** p < 0.05, * p < 0.1; Robust standard errors in parentheses; Standard errors are clustered at district level; Other controls include household head's characteristics (age, gender, educational level, household size, and type of rooftop) and district characteristics (average precipitation and population density).

Table A8

Robustness tests – alternative measures of MEPI.

Dependent variable:	(1)	(2)
	MEPI status	MEPI score
<i>Panel A: Alternative distribution of weights</i>		
Temperature shock	0.022** (0.010)	0.011* (0.006)
Observations	36,242	36,242
R-squared	0.253	0.362
<i>Panel B: Cut-off = 0.5</i>		
Temperature shock	0.025*** (0.009)	0.012** (0.005)
Observations	36,242	36,242
R-squared	0.214	0.342
Other controls	Yes	Yes
Province FE	Yes	Yes
Year FE	Yes	Yes
Province*linear time trend	Yes	Yes
Survey time FE	Yes	Yes

Notes: *** p < 0.01, ** p < 0.05, * p < 0.1; Robust standard errors in parentheses; Standard errors are clustered at district level; Other controls include household head's characteristics (age, gender, educational level, household size, and type of rooftop) and district characteristics (average precipitation and population density).

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